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#include <stdio.h>
#include <stdlib.h>

struct Node {
    int data;
    struct Node* next;
};

struct Queue {
    struct Node *front, *rear;
};

struct Node* newNode(int k) {
    struct Node* temp = (struct Node*)malloc(sizeof(struct Node));
    temp->data = k;
    temp->next = NULL;
    return temp;
}

struct Queue* createQueue() {
    struct Queue* q = (struct Queue*)malloc(sizeof(struct Queue));
    q->front = q->rear = NULL;
    return q;
}

void enqueue(struct Queue* q, int k) {
    struct Node* temp = newNode(k);

    if (q->rear == NULL) {
        q->front = q->rear = temp;
        return;
    }

    q->rear->next = temp;
    q->rear = temp;
    printf("Enqueued %d\n", k);
}

void dequeue(struct Queue* q) {
    if (q->front == NULL) {
        printf("Queue is empty. Cannot dequeue.\n");
        return;
    }
}

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    struct Node* temp = q->front;
    int data = temp->data;

    q->front = q->front->next;

    if (q->front == NULL)
        q->rear = NULL;

    free(temp);
    printf("Dequeued %d\n", data);
}

int isEmpty(struct Queue* q) {
    return (q->front == NULL);
}

void display(struct Queue* q) {
    if (isEmpty(q)) {
        printf("Queue is empty.\n");
        return;
    }
    struct Node* temp = q->front;
    printf("Queue elements: ");
    while (temp != NULL) {
        printf("%d ", temp->data);
        temp = temp->next;
    }
    printf("\n");
}

int main() {
    struct Queue* q = createQueue();
    int choice, val;

    while (1) {
        printf("\n--- Queue Menu ---\n");
        printf("1. Enqueue\n");
        printf("2. Dequeue\n");
        printf("3. Display\n");
        printf("4. Check if Empty\n");
        printf("5. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch(choice) {
            case 1:
                printf("Enter value to enqueue: ");

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        scanf("%d", &val);
        enqueue(q, val);
        break;
    case 2:
        dequeue(q);
        break;
    case 3:
        display(q);
        break;
    case 4:
        if (isEmpty(q))
            printf("Queue is empty.\n");
        else
            printf("Queue is not empty.\n");
        break;
    case 5:
        printf("Exiting...\n");
        exit(0);
    default:
        printf("Invalid choice! Please try again.\n");
}
}
return 0;
}

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